

Rolling Window Cointegration

Mathematical Notes & Journal

`rolling_cointegration.py` — Dark Wing Project

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1 The Problem with Static Cointegration

Both the Engle-Granger (`engle_granger.py`) and Johansen (`johansen.py`) tests estimate a *single, fixed* cointegrating vector $\hat{\beta}$ over the entire sample.

This is mathematically convenient but financially dangerous. Markets are non-stationary in *structure*, not just in price levels. Consider:

- A sector-wide shock (e.g., a budget that hits only one stock)
- A regulatory change (e.g., new FPI rules affecting IT but not energy)
- Fundamental valuation divergence (e.g., one company grows earnings while the other stagnates)

These events can **break the cointegration relationship permanently or temporarily**. A static model will keep signalling trades on a broken pair, leading to sustained losses.

Rolling cointegration is the solution: re-test cointegration on a moving window of recent data so the model *adapts* to structural changes.

2 Mathematical Setup

2.1 The Rolling Window

Let $\{X_t\}_{t=1}^T$ be the full dataset of length T .

Define a **window** of size w observations. At each position $n \in \{w, w+1, \dots, T\}$, we extract the sub-sample:

$$\mathcal{W}_n = \{X_{n-w+1}, X_{n-w+2}, \dots, X_n\} \quad (1)$$

This is the window *ending* at time $t = n$. It contains the most recent w observations. For each window $\mathcal{W}_n$, we estimate:

$$\hat{\beta}_n = \text{OLS slope of } Y \text{ on } X \text{ using data in } \mathcal{W}_n \quad (2)$$

$$\tau_n = \text{ADF } t\text{-statistic on residuals from } \mathcal{W}_n \quad (3)$$

$$p_n = \text{p-value of the ADF test at window } n \quad (4)$$

2.2 Step Size (Efficiency vs. Precision)

Running a full cointegration test at *every* time step (step = 1) is computationally expensive but produces the most granular picture of regime changes.

A **step** $s > 1$ advances the window by s observations at a time:

$$n \in \{w, w+s, w+2s, \dots, T\} \quad (5)$$

Example: Daily data, $w = 252$ (1 year), $s = 5$ (weekly step). This runs the test once per week, giving $\approx (T - 252)/5$ total tests.

For live production code, step = 5 or = 21 (weekly / monthly) is a good balance.

3 The Rolling Engle-Granger

At each window $\mathcal{W}_n$, run the two-step Engle-Granger procedure (see `Engle_Granger_Test.tex`):

$$\text{Step 1: } Y_t = \hat{\alpha}_n + \hat{\beta}_n X_t + \hat{\varepsilon}_t^{(n)} \quad t \in \mathcal{W}_n \quad (6)$$

$$\text{Step 2: } \text{ADF on } \hat{\varepsilon}_t^{(n)} \rightarrow p_n \quad (7)$$

The **output sequence** is a time series of tuples:

$$\{(\hat{\beta}_n, p_n, \mathbb{I}_{p_n < \alpha})\}_{n=w}^T \quad (8)$$

where $\mathbb{I}_{p_n < \alpha}$ is the indicator function (1 if cointegrated, 0 if not) and α is the significance level (e.g., 0.05).

4 The Rolling Johansen

For $m \geq 3$ assets, the rolling procedure re-runs the Johansen eigenvalue problem (see `Johansen_Test.tex`) on each window $\mathcal{W}_n$:

$$\left| \lambda S_{11}^{(n)} - S_{10}^{(n)} (S_{00}^{(n)})^{-1} S_{01}^{(n)} \right| = 0 \quad (9)$$

At each window, we record:

$$r_n^* = \text{cointegrating rank at window } n \quad (10)$$

$$\beta_n^{(j)} = j\text{-th cointegrating vector at window } n \quad (11)$$

Practical note: For rolling Johansen, we fix `k_ar_diff = 1` inside the rolling loop for speed (the optimal lag from the full sample is used as a fixed prior, not recomputed at every window).

5 The Stability Score

The **cointegration stability score** $\mathcal{S}$ quantifies the fraction of the history over which the pair was cointegrated:

$$\mathcal{S} = \frac{1}{N} \sum_{n=1}^N \mathbb{I}_{p_n < \alpha} \quad (12)$$

where N is the total number of rolling windows computed.

Interpretation:

Stability Score	Classification	Action
$\mathcal{S} \geq 0.75$	Highly Stable	Good candidate for systematic pairs trading
$0.50 \leq \mathcal{S} < 0.75$	Moderate	Trade with caution; monitor live
$\mathcal{S} < 0.50$	Unstable	Do not trade algorithmically

6 Rolling Hedge Ratio: Time-Varying $\hat{\beta}_n$

The sequence $\{\hat{\beta}_n\}_{n=w}^T$ is the **rolling hedge ratio**. It tells you how the long-run equilibrium relationship has evolved.

6.1 Why This Matters for Position Sizing

When you construct the spread $S_t = Y_t - \hat{\beta}_n \cdot X_t$, you should use the *most recent* estimate of $\hat{\beta}_n$ rather than the static full-sample $\hat{\beta}$.

Using an outdated $\hat{\beta}$ creates a **stale hedge**: the portfolio is no longer market-neutral, and the "spread" you are trading will have a directional bias.

6.2 Relationship to the Kalman Filter

The rolling hedge ratio $\hat{\beta}_n$ is a *batch* estimate — it re-estimates the hedge ratio from scratch using the last w data points.

The **Kalman filter** (`dynamic_beta.py`) is the online, sequential equivalent: it updates $\hat{\beta}_t$ one data point at a time using a state-space model.

Method	Approach	Best for
Rolling EG (this file)	Batch re-estimation	Regime detection, strategy research
Kalman Filter	Sequential update	Live trading, real-time signal

In the **Dark Wing pipeline**, both are used together:

1. **Rolling cointegration** determines if a pair is worthy of trading (stability score).
2. **Kalman filter** provides the real-time hedge ratio for actual trade execution.

7 Structural Breaks and Regime Classification

A structural break in cointegration occurs when p_n crosses the significance threshold α :

$$\text{Break at time } n \iff p_{n-1} < \alpha \text{ and } p_n \geq \alpha \quad (13)$$

A **recovery** (re-establishment of cointegration) occurs when:

$$\text{Recovery at time } n \iff p_{n-1} \geq \alpha \text{ and } p_n < \alpha \quad (14)$$

In practice: Do not immediately trade a break or recovery signal. Apply a **cooldown rule**: the pair must be continuously cointegrated for at least c consecutive windows (e.g., $c = 10$) before re-entering a position.

8 Window Size Selection

The choice of window size w involves a fundamental bias-variance tradeoff:

Window Size	Advantage	Disadvantage
Small ($w < 100$)	Adapts quickly to regime changes	High variance; unstable $\hat{\beta}$
Large ($w > 500$)	Stable, reliable estimates	Slow to detect breaks
Medium ($w \approx 252$)	Balance of both	Most practical for daily data

Rule of thumb for stocks: $w = 252$ (1 trading year) is the standard for daily data. For intraday data, adjust proportionally (e.g., $w = 252 \times 78$ for 5-minute bars on 252 days).

9 Connection to the Dark Wing Pipeline

- Step 1: Pair Screening:** Run Engle-Granger or Johansen on all candidate pairs using the full historical dataset.
- Step 2: Stability Filter:** Run `RollingCointegration.run()` and compute the stability score $\mathcal{S}$ for each pair.
- Step 3: Shortlist:** Only pairs with $\mathcal{S} \geq 0.75$ are approved for live trading.
- Step 4: Current Status:** Use `RollingCointegration.current_status()` to check if the pair is cointegrated *right now* before placing any order.
- Step 5: Live Execution:** For shortlisted, currently-cointegrated pairs, launch the Kalman filter (`dynamic_beta.py`) to track β_t in real-time.
- Step 6: Monitoring:** Re-run the rolling cointegration weekly (via a scheduler) to detect regime breaks and halt trading automatically.

10 Summary of Key Formulas

$$\text{Window at } n: \mathcal{W}_n = \{X_{n-w+1}, \dots, X_n\} \quad (1)$$

$$\text{Rolling OLS: } \hat{\beta}_n = \frac{\sum_{t \in \mathcal{W}_n} (Y_t - \bar{Y}_n)(X_t - \bar{X}_n)}{\sum_{t \in \mathcal{W}_n} (X_t - \bar{X}_n)^2} \quad (2)$$

$$\text{ADF on residuals: } \Delta \hat{\epsilon}_t = \gamma \hat{\epsilon}_{t-1} + \sum_j c_j \Delta \hat{\epsilon}_{t-j} + u_t \quad (7)$$

$$\text{Stability Score: } \mathcal{S} = \frac{1}{N} \sum_{n=1}^N \mathbb{1}_{p_n < \alpha} \quad (12)$$

$$\text{Structural Break: } p_{n-1} < \alpha \text{ and } p_n \geq \alpha \quad (13)$$